CPSC 304 Project Cover Page

Milestone #: 3
Date: 29th October, 2025
Group Number: 30

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)
In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Project Summary

We are building a relational database for a role-playing game to store users, progress, achievements, locations, items, and characters. The system supports modern game backends and includes a full GUI for managing data and viewing stats—no direct SQL from end users.

Milestones 4 & 5: Plan

Scope:

- SQL init script (DROP/CREATE/INSERT)
- 10 required queries (covering CRUD, selection/projection, joins, aggregations incl. HAVING & nested, and division)
- Full GUI coverage, input sanitization, error handling, documentation, and a live demo.

Phase 1: Database Setup (Oct 30–Nov 3)

- **SQL Init (Tables & Constraints):** Ketan - Nov 2
- **Sample Data & Coverage for Queries:** Samarth - Nov 3
  Outputs: Working DROP/CREATE and INSERTs with meaningful, non-trivial data.

Phase 2: Core GUI & Basics (Nov 3–Nov 13)

- **Main GUI Framework & Navigation:** Norinne, Nov 7
- **Key Operations:**
  - Add User + Save File (INSERT): Ketan - Nov 10
  - Delete User with Cascade (DELETE): Samarth - Nov 10
  - Update Item (UPDATE): Norinne - Nov 12
    Result: Usable app shell with basic create/delete/update flows and clear navigation.

Phase 3: Search & Reports (Nov 12–Nov 20)

- **Character Search (SELECTION):** Ketan - Nov 15
  - Create Character Search page
  - Implement filters: Character type, HP range, Damage range, Location
  - Implement dynamic SELECT query builder with multiple WHERE conditions

- **Custom User Report (PROJECTION):** Norinne - Nov 15
  - Create User Report generator, allow user to select any combination of attributes

- **User Achievements × Objectives (JOIN):** Samarth - Nov 17
  - Create Achievement Tracker page with username search/filter
  - Implement JOIN query across Earned, Achievements_monitors, Objectives, and Users tables

  Result: Flexible filtering, column selection, and achievement linkage.

Phase 4: Aggregations & Complex Queries (Nov 17–Nov 28)

- **Items per Rarity (GROUP BY):** Ketan - Nov 20
  - Create Item statistics page with view button to view items by rarity
  - Implement query with GROUP BY item_rarity

- **Active Users (HAVING):** Norinne - Nov 20
  - Display results showing Username, Number of achievements, Number of objectives completed
  - Filter users with achievement count > threshold

- **Save File Analysis (Nested Aggregation):** Samarth - Dec 23
  - Create Save File Analytics page with Difficulty Analysis button
  - Implement nested aggregation query with outer GROUP BY difficulty and inner subquery for overall average

- **Completionists by Location (Division):** Ketan - Dec 24
  - Create Achievement Completionists page with location selector
  - Implement DIVISION query

  Result: Stats pages for rarity, activity, difficulty, and “all objectives completed” users.

Phase 5: Polish & Delivery (Nov 24–28; Demo prep to Dec 8)

- **Security & Sanitization:** All - Nov 25

- **Error Handling & Notifications:** Samarth - Nov 26

- **Final PDF Documentation:** Norinne - Nov 28

- **Demo Prep (incl. SQL*Plus verification):** All - Dec 8
  Result: Secured app, clear errors/success states, final PDF, and practiced demo.

Risks & Mitigations

- **Availability:** All team members have access to entire codebase. Each member should understand the overall architecture.

- **DB Connectivity:** Multiple members validate Oracle setup.

- **Query Complexity:** Start query implementation early. Each member tests their own queries thoroughly.

- **GUI Integration:** Norinne owns the main framework and will ensure consistent integration. Weekly integration testing on Sundays starting November 3.

- **Time Management:** Early internal deadlines, buffers, and regular check-ins.

Meetings & Communication

- **Weekly:** Mon 6:00 PM (status), Thu 4:00 PM (review), Sun 2:00 PM (integration from Nov 10).

- **Daily:** Group chat; descriptive commits referencing task IDs.

Work Split

All members contribute ~33.3%.

- **Ketan:** Init script; INSERT, SELECTION, GROUP BY, DIVISION; security.

- **Samarth:** DELETE, JOIN, NESTED AGGREGATION; error handling.
- **Norinne:** UPDATE, HAVING; documentation; main GUI.
  Overall load balanced by setup, framework, and documentation ownership.

GUI Expectations (Essentials)

- End-to-end, point-and-click GUI (no text UI, no SQL typing).
- Consistent styling; each query has its own page/controls; results visible in-app; verification via SQL*Plus during demo.

Post-Demo Contact

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